



TREMORWATCH

AI-powered seismic risk intelligence for disaster management & emergency response

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Theme: Disaster Management & Emergency Response Platform

ACM Student Chapter • SVKM IOT Dhule

01 THE PROBLEM

Earthquakes cannot be predicted with certainty — but their **where** and **likelihood** are strongly patterned. When a significant event strikes, every minute of the response gap costs lives and property.

- **Fast triage** — which detected events deserve an immediate response?
- **Risk zones** — where should teams & supplies be pre-positioned?
- **Situational awareness** — what happened, where, and how severe?

02 LIVE DATA

USGS global feed — 13 Aug → 12 Sept 2026 (public GeoJSON, no API key)

11,143

earthquakes in 30 days, worldwide

M 7.8

largest — NNW of Ende, Indonesia

189

significant events ($M \geq 5.0$)

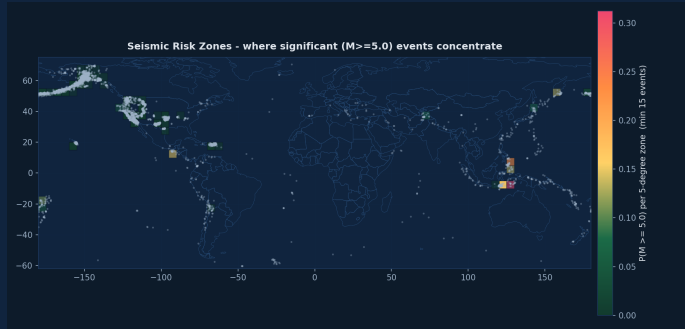
7

tsunamis triggered

03 ANALYSIS PIPELINE

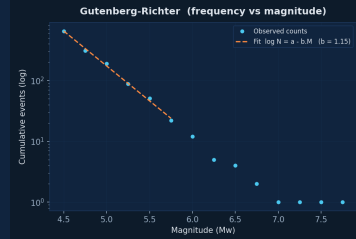
- 1 **INGEST**
Stream every USGS event (location, depth, time, magnitude) as detected
- 2 **EXPLORE**
EDA — Gutenberg-Richter law, depth signatures, rates, hotspots
- 3 **ZONE**
5° grid risk map: observed P(significant) for every zone
- 4 **TRIAGE**
Random Forest scores each event's probability of being $M \geq 5.0$
- 5 **RESPOND**
Ranked alerts drive dispatch, sheltering & resource planning

04 SEISMIC RISK ZONES — THE HEADLINE



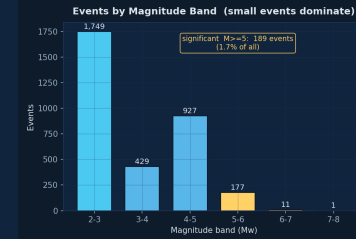
Red = ready. Significant events concentrate in a few zones: **Indonesia / Banda Sea** (333 events, 54 significant), **Timor Leste** ($P = 0.31$), **Philippines** ($P = 0.23$) and **Mexico** ($P = 0.15$) — pre-position response here.

05 FREQUENCY LAW



$b = 1.15 \approx 1.0$ — the live data obeys the Gutenberg-Richter law, the fundamental frequency-magnitude rule of seismology.

06 MAGNITUDE MIX



Small events dominate: only **189 of 11,143 (1.7%)** are significant ($M \geq 5.0$) — yet those drive the response.

07 ML RISK MODEL

Random Forest • features known at detection: location, depth, hour, day • walk-forward CV on 8,912 unseen future events • zero leakage • threshold 0.30

13.3×
LIFT VS
BASELINE

19.6%
PRECISION

78.6%
RECALL

0.314
F1 SCORE

Location (grid) **75.9%**
Depth **16.0%**
Hour of day **5.3%**
Day of week **2.7%**

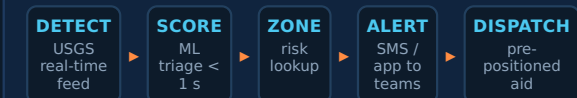
Baseline (alert everything): **1.5%** precision — the signal is almost entirely **location + depth**: the model learned the Ring of Fire on its own.

08 MODEL'S TOP-8 FUTURE ALERTS

DETECTED	LOCATION	M	P(SIG)	HIT
03 Sep	Ende, Indonesia	4.8	0.97	—
03 Sep	Jiangyou, China	5.4	0.95	✓ $M \geq 5$
04 Sep	South Sandwich Is.	5.0	0.95	✓ $M \geq 5$
12 Sep	South Sandwich Is.	5.1	0.94	✓ $M \geq 5$
22 Aug	Ruteng, Indonesia	4.5	0.94	—
04 Sep	Ruteng, Indonesia	4.5	0.94	—
09 Sep	Isangel, Vanuatu	4.7	0.94	—
30 Aug	Sola, Vanuatu	4.8	0.94	—

3 of 8 top-confidence alerts verified significant (37.5% vs **1.5%** base rate) — **25× better than random**; misses cluster just below $M 5.0$.

09 RESPONSE PLATFORM (DESIGN)



Only top-confidence, in-risk-zone events page responders — less alert noise, 79% catch rate.

- **Gutenberg-Richter $b = 1.15$** — live data follows the fundamental seismology law ($b \approx 1$), validating our catalog

- **Deep quakes (>300 km) average M 4.5 vs M 1.6 shallow** — only large deep events are detectable

- **Location + depth = 90% of model signal** — the algorithm learned the Ring of Fire on its own, from data alone

- **13.3× lift** — flagged events are ~13× more likely to be significant; 79% of future significant events caught

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